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DO THANH DAT

**COMPONENT-WISE ANALYSIS OF AN RNA
THREE-DIMENSIONAL STRUCTURE
PREDICTION PIPELINE UNDER
THE TM-SCORE OBJECTIVE**

*Controlled Evaluation of Template Retrieval, Candidate-Source
Diversification, and Geometric Refinement*

MASTER THESIS

DATA SCIENCE

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HANOI, 2026

Abstract

The Stanford RNA 3D Folding competition asks a system to map an RNA sequence to at most five three-dimensional structures and evaluates the best structure using TM-score. This thesis uses that clearly defined objective as an experimental setting for a broader question: how does each major decision in a competitive RNA 3D pipeline affect final structural prediction quality? The starting point is the publicly released pipeline of John, a member of the winning team. The public method is treated as an inherited baseline, not as an exact reconstruction of the winning submission. John and the thesis pipeline are first placed under common data, temporal, exclusion, candidate-budget, and evaluation rules. Groups of components are then changed in a controlled sequence: template-based modelling, candidate-source allocation, and geometric refinement.

Three primary hypotheses were preregistered. H1 compares the retained thesis TBM with J-controlled TBM using five raw candidates. H2 compares five template candidates with three template and two direct DRfold2 candidates while keeping the budget fixed at five. H3 compares Geometry Refinement with a simple source-and-backbone smoother on the same raw candidates. H1 and H2 use best-of-five TM-score. H3 uses sliding-window RMSD over nine residues, or SW-RMSD9, as the primary local metric and requires a predeclared TM-score preservation margin. RNA targets are the statistical units, and regenerated MMseqs sequence-similarity clusters are the dependence blocks.

Development experiments on 20 exposed RNAs selected the final components but did not establish generalization. The frozen method was therefore evaluated once on 97 confirmatory RNAs in 86 sequence-similarity clusters. H1 failed: retained thesis TBM reduced mean TM-score by 0.0255 relative to J-controlled TBM, a 6.358% decrease, with a cluster-bootstrap 95% confidence interval of $[-0.0471, -0.0072]$. H2 passed: replacing two of five template slots with direct DRfold2 candidates increased mean TM-score by 0.0342, or 9.098%, with interval $[0.0155, 0.0543]$. The mixed bank also had lower self-similarity and fewer near-duplicate pairs, supporting source complementarity rather than DRfold2-only superiority.

H3 also passed. Selected Geometry reduced SW-RMSD9 by 0.0205 Å relative to Simple, or 0.473%, with interval [0.0141, 0.0266]. It also increased bank-level TM-score by 0.00218, or 0.532%, and passed the preregistered -0.005 preservation margin. Supporting metrics were mixed: SW-RMSD15 improved, global C1' RMSD and candidate TM were inconclusive, and C1'-IDDT decreased. The final contribution is therefore specific. Fixed-budget candidate-source diversification provides the main confirmed TM-score gain, while Geometry provides a smaller confirmed SW-RMSD9 improvement beyond simple smoothing. The thesis TBM replacement is a negative result and is not claimed as an improvement over the inherited controlled baseline.

Keywords: RNA three-dimensional structure, template-based modelling, DRfold2, geometric refinement, TM-score, SW-RMSD, temporal leakage.

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Chapter 1

Introduction

1.1 Background and motivation

RNA is not only a sequence of A, U, G, and C. The chain folds into motifs and three-dimensional interactions that support regulation, catalysis, and molecular recognition [2, 3, 4]. Experimental methods such as X-ray crystallography, nuclear magnetic resonance, and cryogenic electron microscopy provide high-value reference structures. However, experimental structure determination remains costly and technically demanding, and many RNAs do not have a known close structural relative [1, 5]. Prediction from sequence is therefore both a scientific and an engineering problem.

An RNA 3D prediction system can obtain information from several sources. Template-based modelling, abbreviated TBM, finds an RNA with a known structure and transfers its geometry to the query. A pretrained model learns patterns from existing data and can produce structures even when direct template evidence is weak. A refiner receives an existing structure and adjusts its coordinates to reduce local irregularities. These stages solve different problems. A single end-to-end score does not identify which stage caused a change.

The Stanford RNA 3D Folding competition offers a useful experimental setting. For each target, a system receives the sequence and returns at most five predictions. The score is the highest TM-score across the five predictions and all accepted native conformations. This objective rewards both the accuracy of an individual candidate and the coverage of the complete candidate bank. An independent candidate can add value even when its average quality is lower, provided that it solves a target missed by the other source.

1.2 Research problem

John’s public notebooks provide an important competitive baseline associated with the winning team [28, 29]. Nevertheless, subtracting the score of one public notebook from another submission cannot attribute a gain to one component. The systems may differ simultaneously in their template library, temporal filtering, self-exclusion, pretrained model, fallback, and refinement. The Kaggle leaderboard also provides only an aggregate score on hidden targets rather than paired per-target evidence.

This thesis addresses the attribution problem through controlled decomposition. John and the thesis pipeline are first placed on the same target set, template snapshot, temporal and self-exclusion rules, candidate budget, and evaluator. Each main experiment then replaces one coherent group of components: TBM, candidate-source allocation, or refinement. Local controlled experiments are used for attribution. Kaggle is reserved for external evidence about the complete frozen system.

The study also separates four types of benefit. An accuracy contribution changes native structural accuracy. An availability contribution produces candidates where a method previously failed. A diversity contribution reduces redundant hypotheses in a fixed candidate bank. An engineering safeguard reduces leakage or execution risk without necessarily increasing accuracy. These categories prevent a robustness feature from being reported as a demonstrated score improvement.

1.3 Research questions

The overarching question is:

How do the major stages of an RNA 3D structure prediction pipeline contribute to the TM-score objective used in the Kaggle competition?

Three specific questions define the experimental sequence:

- RQ1.** Which changes to the inherited TBM pipeline affect best-of-five TM-score under a common controlled protocol, and what roles do retrieval, ranking, and reconstruction play?
- RQ2.** Given a fixed budget of five predictions, does replacing two template candidates with candidates from an independent pretrained predictor improve structural coverage and best-of-five TM-score?

RQ3. Can Geometry Refinement improve local C1' structural accuracy beyond simple smoothing while preserving the global fold measured by TM-score?

1.4 Contributions and scope

The thesis does not claim that the full pipeline was invented from scratch. TBM and the hybrid direction are inherited from John's public work. The scientific contribution is a controlled, auditable analysis of changes made to that baseline. Each claim is tied to the experiment that can support it. A component is described as a demonstrated contribution only when it passes its preregistered gate. A component that only reduces redundancy or prevents leakage is described as a safeguard.

The study contributes a 5,135-row data provenance ledger, an exact reproduction of the production geometry prior, a shared controlled template database, a partial but executable reconstruction of the public John pipeline, fixed-budget source-allocation experiments, and a same-candidate refinement factorial. It also provides deterministic cluster-aware inference and a final pre-opening receipt that freezes the target manifest, code, data, checkpoints, priors, fallback rules, metrics, and statistics.

The local representation contains one C1' coordinate per nucleotide. The main competition objective is best-of-five TM-score. Local refinement is assessed with SW-RMSD9 and supporting metrics. Diagnostics directly optimized by Geometry, including angle likelihood, torsion likelihood, clash count, kink count, and radius-of-gyration deviation, are not treated as independent proof of native accuracy.

Chapter 2

Background and Problem Definition

2.1 RNA coordinate representation

Each sequence record has a target identifier, an RNA sequence, a temporal cutoff, and descriptive metadata. Each label record represents one residue and contains its base, position, and three-dimensional coordinates. The competition uses the sugar atom C1' as the representative point for each nucleotide. A prediction for an RNA of length L is therefore a matrix

$$X \in \mathbb{R}^{L \times 3}, \quad (2.1)$$

where row i stores (x_i, y_i, z_i) for residue i . A target may have several valid experimental conformations. The evaluator must therefore compare the candidate bank with all accepted references.

2.2 TM-score and best-of-five evaluation

TM-score measures structural similarity after finding a suitable rotation and translation. In a general form,

$$\text{TM}(X, Y) = \max_{\mathcal{A}} \frac{1}{L_{\text{ref}}} \sum_{i=1}^{L_{\text{ali}}} \frac{1}{1 + (d_i/d_0(L_{\text{ref}}))^2}, \quad (2.2)$$

where d_i is the distance between an aligned pair of C1' atoms after superposition, and d_0 depends on reference length [22, 23]. A close pair contributes a value near one. The contribution decreases smoothly as the distance increases. Unlike ordinary RMSD, a small number of large local errors do not dominate the entire score through a squared penalty. TM-

score therefore emphasizes the global fold.

If target q has five predictions $X_q^{(1)}, \dots, X_q^{(5)}$ and R_q accepted native conformations, the target score is

$$S_q = \max_{k \in \{1, \dots, 5\}} \max_{r \in \{1, \dots, R_q\}} \text{TM}(X_q^{(k)}, Y_q^{(r)}) . \quad (2.3)$$

For example, if the best native TM-scores of the five predictions are 0.31, 0.42, 0.38, 0.57, and 0.44, the target receives 0.57. A second candidate that closely duplicates the first may add no value. A candidate from another source can increase the score if it covers a different plausible fold. This is why candidate count and candidate diversity must be controlled when source contribution is measured.

2.3 Template-based modelling

TBM is a sequence of algorithms rather than a single copying step. Retrieval finds known RNA chains that may be related to the query. Alignment maps query residues to template residues. Ranking estimates which mappings are most useful. Coordinate transfer copies resolved C1' positions. Gap handling builds positions for query residues without a template counterpart. Candidate selection finally allocates the limited prediction slots.

John's public TBM performs an exhaustive scan after a length filter. It computes global alignment, local alignment, similarity between hand-crafted sequence features, and the Jaccard overlap of 3-mers, with weights 0.4, 0.3, 0.2, and 0.1. A k -mer is every contiguous substring of length k . For sequence 'AUGCA', its 3-mers are 'AUG', 'UGC', and 'GCA'. Jaccard similarity divides the number of distinct shared 3-mers by the size of their union. Exhaustive search means that the score is computed for every template remaining after the filter rather than only for a shortlist returned by a search engine.

2.4 Pretrained prediction and candidate diversity

DRfold2 is a pretrained RNA structure predictor that combines learned sequence representations and learned geometric information [16, 18]. In this thesis, T denotes a raw template candidate and D denotes a direct DRfold2 candidate. Bank $5T$ assigns all five slots to templates. Bank $3T + 2D$ replaces two template slots with DRfold2 while preserving $N = 5$.

Candidate diversity is measured without native labels using pairwise self-TM, the TM-score between candidates within the same bank. A high self-TM indicates similar hypotheses.

A lower self-TM indicates wider fold coverage, although it does not prove that any candidate is correct. Near-duplicate pairs are defined using a self-TM threshold of 0.95. These measures explain a possible mechanism but cannot replace the native best-of-five score.

2.5 Geometric refinement and local accuracy

A refiner does not generate a fold from nothing. It receives raw coordinates and moves each C1' point to reduce an objective. The Simple baseline anchors the output to the source through a Huber loss and smooths adjacent backbone distances. Geometry Refinement adds collision avoidance, empirical pseudo-angle and pseudo-torsion priors, and a guard against newly introduced sharp kinks. A source anchor is essential because the candidate-generation stage may already contain a useful global fold.

SW-RMSD9 evaluates overlapping windows of nine residues. Each predicted window is superposed on the corresponding native window before RMSD is computed. It therefore asks whether a short segment has the correct local shape. C1'-IDDT compares internal distances under thresholds of 0.5, 1, 2, and 4 Å within a local radius [24]. SW-RMSD15, global C1' RMSD, and C1'-IDDT are supporting metrics.

Angle negative log-likelihood, torsion negative log-likelihood, clash proxies, kink counts, pair-like fractions, and radius-of-gyration deviations are useful diagnostics. However, Geometry directly optimizes several of them. A lower optimization diagnostic only proves that the optimizer reduced its own objective. Independent native-based metrics are required to support an accuracy claim.

Chapter 3

Data, Provenance, and Leakage Control

3.1 Kaggle train V2 and data roles

The Kaggle ‘train_sequences.v2.csv’ file contains exactly 5,135 RNA records. The V4 master ledger has one row per record and stores the temporal cutoff, sequence length, normalized sequence hash, native technical validity, prior-training use, development exposure, exact-duplicate status, regenerated MMseqs sequence-similarity cluster, model-overlap audit, external-exposure audit, and final inclusion state.

The data flow assigns different roles before performance is inspected. The 3,397 RNAs with temporal cutoffs before 27 May 2022 reproduce ‘P0-production’, the deployed distance, size, angle, and torsion priors. These RNAs are prior-training data rather than final-test data. Phase 1 rebuilt both production JSON artifacts from the exact 3,397-target manifest. Every numeric value matched the deployed artifact, with maximum absolute error zero at tolerance 10^{-10} .

The later branch begins with 878 RNAs after the declared structural cutoff. Restricting length to 30 through 400 leaves 525. Requiring a canonical sequence and a technically valid native C1’ structure leaves 408. Exact-sequence deduplication retains 315 representatives. MMseqs clustering at 80% identity, 80% coverage, and coverage mode zero creates 227 sequence-similarity clusters. Removing 90 clusters with known development exposure leaves 172 provisional RNAs in 137 clusters.

DRfold2 structural-training overlap screening finds 19 exact overlaps and 75 targets with an MMseqs homolog at the same 80% identity and 80% coverage threshold. Ninety-seven targets pass this structural-overlap screen. The accessible repository and external text audit finds no additional exposure evidence for these targets. All 97 targets in 86 clusters are retained without sampling in the final confirmatory manifest.

3.2 Development exposure

The exposure ledger conservatively marks any target whose native performance was scored, whose per-target result was inspected, or whose result influenced a method, hyperparameter, source weight, failure analysis, table, or plot. The ledger includes the 3,397 prior-training RNAs, 107 additional exposed train-V2 targets, and 12 CASP15 targets managed in a separate development ledger. An exposed target and its sequence-similarity cluster cannot enter the confirmatory set.

The 20-target V4 development cohort comes from a historical calibration split and was already exposed. It is suitable for component selection but never for confirmatory claims. The cohort contains 17 regenerated MMseqs clusters. The term “sequence-similarity cluster” is used deliberately. MMseqs similarity alone does not establish a biological family or a shared function.

3.3 Temporal filtering and self-exclusion

A template is allowed only when its release date is strictly earlier than the target cutoff. Every PDB associated with the target record is excluded from retrieval. The same filter is applied to John and thesis TBM under the controlled protocol. This prevents a pipeline from retrieving the answer or a structure that became available only after the prediction date.

An intentionally unsafe diagnostic is retained only to measure leakage sensitivity. It allows later templates while keeping the other operations fixed. A higher unsafe score is evidence that temporal leakage can inflate evaluation. It is not a deployable method and is never interpreted as an accuracy contribution.

3.4 Pretrained-model provenance

Structural training and sequence-language-model pretraining are treated as separate provenance streams. The audited DRfold2 source is fixed at commit ‘3667d6a’, and the cfg97 branch uses 20 checkpoint files recorded by SHA-256. The local structural FASTA contains 10,432 records and 3,414 unique sequences. The relation between that file and the public archive is not proven byte-exact.

DRfold2 reports that its RCLM sequence model was trained for 67,000 batches on roughly 30 million sequences from RNAcentral Release 22 [18]. No complete corpus membership manifest is locally available. The thesis therefore does not describe the whole pre-

trained model as time-safe. The final pool passes the structural-training overlap screen, while unresolved language-model membership remains a limitation.

Boltz cannot be used as primary V4 evidence because the required checkpoint, exact source state, and training provenance are unavailable. The public John hybrid route that depends on Boltz is consequently only partially reproducible. This limitation does not invalidate the controlled TBM comparator or the direct DRfold2 allocation experiment.

Chapter 4

Methods

4.1 Public John baseline and controlled variants

The baseline is named the reproduced publicly released John pipeline. It is not called the exact winning pipeline because the public notebook, original attached data, dependency versions, and final competition submission have not been shown to be identical. Three labels are kept separate:

- ‘J-reported’ denotes scores stated on Kaggle or by John and is contextual only.
- ‘J-original-local’ denotes the captured public code under the locally available artifacts.
- ‘J-controlled’ denotes John’s algorithm under the same database, temporal, exclusion, target, candidate-count, and evaluator rules as the thesis.

The public TBM computes the four-part composite score, transfers coordinates through global alignment, completes gaps, applies a rule-based refiner, and adds jitter. H1 must isolate candidate generation, so its raw boundary is after coordinate transfer and gap completion but before rule refinement and jitter. ‘J-controlled-TBM-raw’ is taken at this boundary. A SHA-256-derived seed replaces the public use of Python’s process-randomized hash for deterministic fallback generation. This is documented as an execution reproducibility change rather than a scientific modification.

The public hybrid notebook routes an entire target to template outputs or to a Boltz-conditioned DRfold2 branch, then duplicates or falls back when necessary. The control-flow and fallback behavior are executable locally, but the Boltz-conditioned structural output is unavailable. J-controlled full hybrid is therefore descriptive and cannot be used to attribute a component gain.

4.2 Common template database

The controlled template database contains 23,869 valid parsed RNA chains from 8,613 PDB entries. Each chain has a sequence, release date, resolved-residue mask, and C1' coordinate array. Metadata, coordinates, and per-chain manifests have separate hashes. John and thesis methods read the exact same snapshot. A target-specific filter then applies the cutoff date, PDB self-exclusion, and length compatibility.

The length filter uses the absolute length difference divided by the larger length. A ratio at most 0.6 is allowed when either sequence is shorter than 50 residues, at most 0.2 when either exceeds 1,000, and at most 0.4 otherwise. The relaxed short-RNA rule avoids discarding partial but useful short templates, whereas the long-RNA rule limits expensive and implausible comparisons.

4.3 Retained thesis TBM

The final retained TBM is a specific five-stage algorithm selected from the development ablations.

1. **Eligibility filtering.** Remove templates that fail the length rule, have a release date on or after the target cutoff, or belong to a PDB associated with the target.
2. **Global exhaustive retrieval.** Compute the global alignment component for every eligible chain and retain the 50 highest-scoring chains. Ties are broken by the stable chain identifier.
3. **Alignment and ranking.** Realign each shortlisted chain. Identity is the fraction of identical nucleotides in aligned non-gap columns. Query coverage is the fraction of target residues that receive a finite template coordinate. Rank by identity multiplied by query coverage.
4. **Distinct-PDB selection.** Select the highest-ranked chain from each unused PDB until five PDBs are represented or the distinct pool is exhausted. Then backfill from the remaining ranked chains.
5. **Coordinate transfer and gap completion.** Copy finite C1' coordinates through the alignment. Fill internal missing spans by linear interpolation between the two known endpoints. Extend terminal gaps along the nearest backbone direction. Use deterministic de novo coordinates if fewer than five candidates remain.

MMseqs, the local-alignment component, hand-crafted features, 3-mer Jaccard, and template completeness are not part of the retained method. They remain in the RQ1 ablation registry so that their measured effects are reported rather than hidden. Identity multiplied by coverage is retained as an engineering heuristic because its development effect is positive but inconclusive. Distinct-PDB selection is retained as a diversity safeguard, not as an accuracy claim.

4.4 Direct DRfold2 candidate generation

The pretrained branch uses direct cfg97 end-to-end outputs from 20 checkpoint files. It deliberately stops before the expensive DRfold2 potential-optimization stage, then converts the two highest mean-pLDDT checkpoint outputs into all-atom structures. The C1' chain, residue confidence, global mean confidence, and optional learned distance priors are normalized into a pickle-free candidate contract.

The primary bank comparison uses the same retained thesis template list. $5T$ takes the first five template candidates. $3T + 2D$ takes the first three template candidates and two direct DRfold2 candidates. If a DRfold2 slot fails, the next frozen template candidate fills that slot and the target remains in the all-target analysis. The failure is still recorded as a DRfold2 source failure.

The mechanism banks $2T$, $1T + 1D$, and $2D$ help distinguish source strength from source complementarity. They do not replace the primary fixed- N comparison. Boltz is excluded because its required artifacts are unavailable.

4.5 Simple smoothing baseline

Simple starts from raw coordinates X_0 and optimizes coordinates X for 300 Adam steps with learning rate 0.04. Its objective contains a source Huber term and an adjacent-backbone Huber term:

$$\mathcal{L}_{\text{Simple}} = 3.0 \mathcal{L}_{\text{source}} + 1.0 \mathcal{L}_{\text{backbone}}. \quad (4.1)$$

The source term is confidence weighted and prevents the optimizer from rewriting the candidate. The backbone term penalizes standardized deviations from the P0 adjacent C1' distance. Both use Huber loss so that a small number of severe gaps provide a repair gradient without dominating every other residue. Simple uses fixed geometry strength one

and disables clash, size, angle, torsion, kink, and pair-context terms.

4.6 Selected Geometry Refinement

Geometry uses the same source and backbone foundation and adds clash, global pseudo-angle, global pseudo-torsion, and kink-regression terms:

$$\mathcal{L}_{\text{Geometry}} = 3.0 \mathcal{L}_{\text{source}} + s(c) (1.0 \mathcal{L}_{\text{backbone}} + 0.3 \mathcal{L}_{\text{clash}} + 0.3 \mathcal{L}_{\text{angle}} + 0.15 \mathcal{L}_{\text{torsion}} + 20.0 \mathcal{L}_{\text{kink}}), \quad (4.2)$$

where

$$s(c) = 0.2 + 0.8(1 - c) \quad (4.3)$$

and c is the candidate’s global confidence clipped to $[0, 1]$. Low-confidence candidates receive stronger geometric correction, while high-confidence candidates remain more strongly anchored to the source. The optimizer runs for 300 Adam steps at learning rate 0.04 with gradient clipping at 10.

The angle and torsion priors are unconditional global P0 distributions. A candidate-derived pair-like context was tested but did not beat this simpler control. The radius-of-gyration term is disabled because it did not show a clear independent benefit. Fixed geometry strength produced a better SW-RMSD9 point estimate, but its TM confidence interval relative to production Geometry reached -0.00526 , just beyond the preregistered -0.005 preservation margin. Adaptive strength was therefore kept.

The kink term does not force every raw sharp angle toward a universal value. It penalizes a newly introduced or worsened kink relative to the raw source. This limits local regression without assuming that every sharp native-like motif is invalid.

4.7 Failure handling

Missing output is never silently removed. A missing template slot uses a deterministic method-specific de novo fallback. A missing DRfold2 slot uses the next frozen template while preserving the source-failure flag. A numerical failure in Simple or Geometry returns the corresponding Raw candidate and records the exception. If a complete bank still fails

after every fallback, its primary target TM-score is zero. The primary population therefore remains the complete frozen manifest.

Chapter 5

Experimental Design and Statistics

5.1 Primary hypotheses

H1 compares retained thesis TBM with J-controlled TBM on the same targets, controlled database, temporal and self-exclusion rules, evaluator, and five raw candidates. The paired effect is

$$\Delta_{H1} = \text{TM}_{\text{Thesis TBM}} - \text{TM}_{\text{J-controlled TBM}}. \quad (5.1)$$

H2 isolates candidate-source allocation. It compares $3T + 2D$ with $5T$ using the same retained thesis TBM, the same targets, five raw candidates, and no refinement:

$$\Delta_{H2} = \text{TM}_{3T+2D} - \text{TM}_{5T}. \quad (5.2)$$

H3 compares selected Geometry with Simple on the same raw $3T + 2D$ candidates. For each target, SW-RMSD9 is first averaged over the five paired candidates. The superiority effect is written in the positive direction:

$$\Delta_{H3} = \text{SW9}_{\text{Simple}} - \text{SW9}_{\text{Geometry}}. \quad (5.3)$$

H3 also requires the lower bound of the 95% cluster-bootstrap interval for $\text{TM}_{\text{Geometry}} - \text{TM}_{\text{Simple}}$ to be greater than -0.005 . If H3 fails either the local superiority gate or the TM preservation gate, Simple becomes the final refiner.

5.2 Candidate-level and bank-level evaluation

Candidate-level evaluation chooses the native conformation that gives the raw candidate its highest TM-score. Raw, John, Simple, and Geometry are then compared with that same locked conformation. A refiner is not allowed to select a different native reference that favors its own output. Candidate metrics are averaged within target before targets are aggregated.

Bank-level evaluation follows the competition rule and takes the highest TM-score over the five candidates and every accepted native conformation. Candidate rows are not treated as independent samples. The statistical unit is always the RNA target.

5.3 Cluster-aware inference

Each primary hypothesis produces one paired delta per target. The point estimate is the target-weighted arithmetic mean. A cluster bootstrap samples the K regenerated MMseqs sequence-similarity clusters with replacement 10,000 times using NumPy PCG64 seed 20260819. Whenever a cluster is sampled, all member targets are carried with it. The 2.5th and 97.5th percentiles form the confidence interval.

The one-sided raw p -value uses 100,000 cluster sign flips with the same seed. Every target in a cluster receives the same random sign. The statistic is the target-weighted mean delta. H1, H2, and H3 form one Holm step-down family at family-wise alpha 0.05. The H3 TM preservation interval is a separate noninferiority safeguard outside the Holm family.

Every result reports the number of targets, number of clusters, mean and median delta, cluster-bootstrap interval, raw p -value, Holm-adjusted p -value, and counts of improved, tied, and regressed targets. An absolute delta below 10^{-6} is a tie. Scores are displayed to three decimal places, while small deltas and intervals retain four or more decimal places. All decisions use unrounded values.

5.4 Headline factorial

Table 5.1: Same-candidate refinement factorial.

Candidate bank	Raw	John original	John fixed	Simple	Geometry
J-controlled candidates	✓	✓	✓	✓	✓
Thesis $3T + 2D$ candidates	✓	✓	✓	✓	✓

The John-original cell uses the raw candidate confidence under the public rule-based refiner. John-fixed sets rule confidence to 0.5 so that source-specific confidence scales cannot explain the comparison. Simple and Geometry use identical raw coordinates and the exact optimizer definitions above.

Chapter 6

Results

6.1 Phase 1 audits

The P0 manifest contains exactly 3,397 RNA targets. Rebuilt distance and size priors and rebuilt angle and torsion priors match the deployed production artifacts with maximum absolute numeric error zero. This establishes reproduction, not superiority over another prior. No P1 prior was selected.

The public John TBM behavior is executable under the available local artifacts. The hybrid routing, padding, and fallback logic are also executable, but the structural Boltz-conditioned branch cannot be reproduced. John is therefore retained as a partial public-pipeline reproduction. The J-controlled raw TBM comparator remains valid because it uses the shared controlled database and does not depend on Boltz.

The final structural-overlap audit retains 97 of 172 provisional targets. No additional exposure is found in the accessible audit. RCLM corpus membership remains unavailable, and complete pretrained time-safety is not claimed.

6.2 Development RQ1: TBM decomposition

On 20 development targets in 17 clusters, J-controlled TBM reaches mean best-of-five TM 0.389 and retained thesis TBM reaches 0.395. The paired delta is +0.0056, or +1.45% relative to the baseline mean. The 95% cluster-bootstrap interval is $[-0.0030, 0.0170]$, and the raw one-sided p -value is 0.1558. Ten targets improve, two tie, and eight regress. H1 does not pass because the interval crosses zero and the Holm procedure does not reject it. The thesis does not claim a demonstrated TBM accuracy improvement over John.

Table 6.1: Development TBM component effects. A positive delta favors the first method.

Comparison	Mean delta	Raw p	Decision
Retained thesis - J-controlled	+0.0056	0.1558	Positive but inconclusive
MMseqs plus composite - composite	-0.0030	1.0000	Remove MMseqs
Full composite - global-only	-0.0057	0.9729	Retain global-only
Identity $\times$ coverage - identity	+0.0206	0.0985	Retain as heuristic
Completeness beyond coverage	0.0000	1.0000	Remove completeness
Distinct-PDB selection	-0.0013	0.7496	Retain as diversity safeguard
Unsafe dates - time-safe	+0.0315	0.0079	Leakage diagnostic only

Adding MMseqs does not improve availability because both composite configurations already produce complete banks. Full composite is lower than global-only by 0.0057, with an interval entirely below zero. Identity multiplied by coverage has a useful positive point estimate but an interval that crosses zero. Completeness adds exactly zero beyond coverage. Distinct-PDB selection remains within the 0.005 tolerance and is kept only to reduce redundancy.

Curved gap completion does not beat linear completion in any preregistered gap-length bin. Linear is therefore selected as the simplest noninferior method. Allowing unsafe dates increases TM by 0.0315 with interval $[0.0086, 0.0569]$. This result demonstrates the importance of temporal filtering rather than supporting an unsafe method.

6.3 Development RQ2: candidate-source diversification

Table 6.2: Fixed-budget development candidate banks.

Bank	Availability	Mean best-of-five TM	Mean self-TM	Near-duplicate pairs
$5T$	100%	0.395	0.453	8%
$3T + 2D$	100%	0.411	0.380	2%
$2T$	100%	0.381	0.543	0%
$1T + 1D$	100%	0.399	0.296	0%
$2D$	100%	0.288	0.501	0%

The primary $3T + 2D$ versus $5T$ delta is +0.0157, or +3.97% relative to the $5T$ mean. Its interval is $[-0.0056, 0.0419]$, and raw $p = 0.1371$. Eight targets improve, 11 tie, and one regresses. H2 does not pass and is reported as positive but inconclusive. The allocation was

retained in the frozen method because this was the preregistered development decision, not because improvement had already been proven.

Mean self-TM decreases from 0.453 to 0.380, a 16.05% reduction, while the fraction of near-duplicate pairs falls from 8% to 2%. These native-blind measures support the mechanism of wider hypothesis coverage. However, $2D$ reaches only 0.288 compared with 0.381 for $2T$. DRfold2 is not uniformly more accurate than templates. The most defensible mechanism is independent-source complementarity rather than DRfold2-specific superiority.

6.4 Development RQ3: refinement beyond Simple

The production Geometry configuration first reduces SW-RMSD9 relative to Simple by 0.0220 Å, with interval [0.0060, 0.0388]. The mechanism gates then remove candidate-derived context and the radius-of-gyration term. Adaptive strength remains because fixed strength narrowly fails the TM preservation requirement. This combined configuration is frozen before it is scored.

Table 6.3: Same-candidate refinement on the thesis $3T + 2D$ development bank.

Setting	Mean SW-RMSD9 (Å)	Mean best-of-five TM	Role
Raw	4.560	0.411	No refinement
John original	4.473	0.408	Public-rule sensitivity
John fixed	4.444	0.398	Fixed-confidence control
Simple	4.555	0.408	Primary comparator
Selected Geometry	4.532	0.409	Development-selected refiner

Selected Geometry reduces SW-RMSD9 relative to Simple by 0.02345 Å, or 0.515%. The interval is [0.00689, 0.04062], raw $p = 0.00769$, and Holm-adjusted $p = 0.02307$. Fifteen targets improve and five regress. The H3 local-superiority gate passes.

Mean best-of-five TM increases from 0.40805 for Simple to 0.40884 for selected Geometry. The delta is +0.000787, or +0.193%, with interval [−0.00318, 0.00433]. The interval does not demonstrate a TM increase, but its lower bound remains above the −0.005 preservation margin. The safeguard passes. None of the 200 selected-refinement executions fails.

Raw $3T + 2D$ still has mean TM 0.41068, approximately 0.00184 above selected Geometry. The supported claim is therefore narrower than an end-to-end improvement: Geometry improves local native accuracy beyond Simple and preserves TM relative to Simple. It does not demonstrate a TM gain relative to Raw. John fixed has the lowest SW-RMSD9 in the table but reduces mean TM to 0.398. This result illustrates why a local metric cannot replace the global competition objective.

6.5 Development primary family and component decisions

H3 has the smallest raw p -value, 0.00769, and passes the first Holm threshold $0.05/3$. Its adjusted p -value is 0.02307. H2 has raw $p = 0.13712$ and fails the next step. Holm step-down stops, and H1 is not rejected. H3 is the only primary development hypothesis that passes the multiplicity-controlled family.

Table 6.4: Final development component decisions.

Component	Decision	Evidence label
P0-production prior	Keep	Exact reproduction; no P1 selected
MMseqs branch	Remove	No quality or availability gain
Full composite	Remove	Global-only is better on development
Identity $\times$ coverage	Keep	Engineering heuristic, not demonstrated contribution
Distinct PDB	Keep	Diversity safeguard
Linear gap completion	Keep	Simplest method not beaten by curved completion
$3T + 2D$ allocation	Keep in frozen method	Positive but inconclusive development evidence
Boltz	Remove	Required artifacts unavailable
Adaptive strength	Keep	Fixed control fails TM preservation margin
Candidate-derived context	Remove	Does not beat unconditional prior
Radius-of-gyration term	Remove	No clear independent benefit
Selected Geometry	Keep	H3 and TM safeguard pass

6.6 Confirmatory execution and opening receipt

The confirmatory manifest contains 97 targets in 86 MMseqs sequence-similarity clusters, with no sampling. Candidate generation used the frozen commit, database, priors, DRfold2 checkpoints, fallback rules and per-target seeds. Before native labels were introduced, the execution produced and hashed 3,011 scientific files totalling 8.940 GB. The completeness gate found 97 TBM banks, 97 raw banks, 97 refinement factorials, 194 direct DRfold2 candidates and zero generation failures or fallback targets.

An explicit opening receipt then recorded the output-manifest hash, evaluator code, US-align binary, local metrics, cluster-aware statistics and the authorization for one native-label opening. The labels were opened once, all 97 targets were evaluated, and no method or statistical procedure was changed. The evaluator reported zero scoring failures. Development and confirmatory results remain separate in every artifact.

6.7 Confirmatory RQ1: controlled TBM comparison

J-controlled 5T raw candidates achieve mean best-of-five TM 0.401, whereas retained thesis 5T raw candidates achieve 0.376. The paired effect $TM_{\text{Thesis}} - TM_{\text{J-controlled}}$ is -0.0255 , a 6.358% decrease relative to J-controlled. The 95% cluster-bootstrap interval is $[-0.0471, -0.0072]$. The one-sided raw p -value for the preregistered positive direction is 0.995940, and Holm does not reject H1. Forty-three targets improve, three tie and 51 regress.

H1 therefore fails with a negative confirmatory effect. The retained thesis TBM cannot be presented as more accurate than J-controlled TBM. This conclusion is stronger than the development result, which had a small positive point estimate but wide uncertainty. The direction does not generalize.

The preregistered length sensitivity is descriptive. The 30–79 nt stratum is near neutral at $+0.0007$. The 80–149 nt stratum is -0.0453 with an interval below zero. The 150–249 nt and 250–400 nt strata are also negative but less precise. These results help locate the failure but do not replace the all-target H1 test.

6.8 Confirmatory RQ2: fixed-budget source diversification

Table 6.5: Confirmatory fixed-budget candidate-source results. Self-TM and the near-duplicate fraction are native-blind diversity measures.

Bank	Mean best TM	Mean self-TM	Near-duplicate pairs	Availability
$5T$	0.376	0.518	0.149	100%
$3T + 2D$	0.410	0.369	0.054	100%
$2T$	0.348	0.533	0.124	100%
$1T + 1D$	0.392	0.266	0.010	100%
$2D$	0.333	0.466	0.021	100%

Replacing two template slots with two direct DRfold2 candidates increases mean best-of-five TM from 0.376 to 0.410. The paired effect is +0.0342, or 9.098% relative to $5T$, with cluster-bootstrap interval [0.0155, 0.0543]. The raw one-sided p -value is 0.000190 and Holm-adjusted $p = 0.000380$. Forty-seven targets improve, 33 tie and 17 regress. H2 passes.

Mean pairwise self-TM decreases by 28.644%, and the near-duplicate pair fraction decreases by 64.138%. The matched- N results explain why this is a diversity claim. The $2D$ bank scores 0.333 compared with 0.348 for $2T$, so direct DRfold2 is not uniformly stronger in isolation. In contrast, $1T + 1D$ scores 0.392. Combining independent sources covers different structural hypotheses and raises the chance that one of the five predictions is close to the native fold.

6.9 Confirmatory RQ3: Geometry beyond Simple

Selected Geometry reduces mean candidate-level SW-RMSD9 from 4.325 Å to 4.305 Å. The target-level paired reduction is 0.0205 Å, or 0.473%, with cluster-bootstrap interval [0.0141, 0.0266]. The raw one-sided p -value is 0.000010 and Holm-adjusted $p = 0.000030$. Geometry improves 73 targets and regresses 24. The preregistered local-superiority test passes.

At bank level, Geometry increases mean TM from 0.410 for Simple to 0.412. The paired effect is +0.00218, or 0.532%, with interval [0.00097, 0.00366]. This lower bound is above the preregistered noninferiority margin of -0.005 . H3 therefore passes both the local-superiority gate and the global TM safeguard.

Table 6.6: Frozen same-candidate confirmatory refinement factorial. John fixed is the main John cell; John original is retained as a confidence sensitivity.

Candidate bank	Raw	John original	John fixed	Simple	Geometry
J-controlled TM	0.401	0.398	0.393	0.399	0.402
Thesis $3T + 2D$ TM	0.410	0.407	0.404	0.410	0.412

John fixed reduces bank TM by 0.0086 on J-controlled candidates and by 0.0065 on thesis candidates. John original is less damaging but remains below Raw. Simple is approximately neutral relative to Raw on the thesis bank, with effect -0.00040 and an interval crossing zero. Geometry is $+0.00178$ above Raw with interval $[0.00037, 0.00336]$. These secondary comparisons support the same conclusion as the primary Geometry-versus-Simple test without replacing it.

6.10 Supporting metrics and sensitivity

Supporting native metrics do not all move in the same direction. Geometry improves SW-RMSD15 over Simple by $0.0143 \text{ \AA}$, with interval $[0.0087, 0.0200]$. Global C1' RMSD changes by only $0.0028 \text{ \AA}$, with an interval crossing zero. Same-reference candidate TM changes by $+0.00018$ and is also inconclusive. In contrast, C1'-lDDT decreases by 0.00521 , with interval $[-0.00671, -0.00369]$. The H3 claim must therefore remain tied to the preregistered SW-RMSD9 effect and TM safeguard. It is inaccurate to state that Geometry improves every independent local metric.

All 97 targets have two direct DRfold2 outputs and all pass the frozen structural overlap screen, so source-available and complete-case sensitivities equal the primary population. H2 and H3 effects are strongest in the 80–149 nt stratum and remain positive in the larger preregistered bins. The 30–79 nt strata are inconclusive. These analyses describe effect heterogeneity and do not alter the primary decisions.

6.11 Confirmatory primary family

H3 has raw $p = 0.000010$ and Holm-adjusted $p = 0.000030$. H2 has raw $p = 0.000190$ and Holm-adjusted $p = 0.000380$. Both pass the step-down family and have confidence intervals above zero. H1 has raw $p = 0.995940$ in the preregistered positive direction and does not pass. The final decisions are therefore H1 fail, H2 pass and H3 pass, with the H3 TM safeguard also passing.

Chapter 7

Discussion

7.1 What changed relative to the inherited baseline

The public John method supplies the baseline ideas of exhaustive template scoring, coordinate adaptation, rule-based refinement, pretrained candidates, and template fallback. The thesis contribution is not the use of these ideas in isolation. It is the controlled replacement and evaluation of specific groups of decisions under a shared protocol.

The retained thesis TBM is simpler than both the public four-part composite and the initial thesis pipeline. Development evidence removed MMseqs, full composite scoring, completeness and curved gaps. Coverage remains a heuristic, while distinct-PDB selection remains a diversity safeguard. This simplification was a valid development decision, but the confirmatory result shows that the resulting TBM bank is less accurate than J-controlled TBM. Simpler does not mean superior, and no thesis TBM accuracy contribution is claimed.

The candidate bank differs from John’s public hybrid routing. John routes a target mostly to one source, whereas the thesis reserves slots for both sources in every eligible target. The fixed- N experiment estimates this allocation effect without rewarding the mixed bank for having more candidates. This is the largest confirmed TM-score contribution in V4: $3T + 2D$ raises mean TM by 0.0342 relative to $5T$.

The refiner is also different from John’s three-rule procedure. Geometry uses a continuous optimization objective with source anchoring and empirical priors. In the confirmatory same-candidate factorial, Geometry improves SW-RMSD9 beyond Simple and slightly raises bank TM. The effect is much smaller than the source-allocation gain, and the unfavorable IDDT result prevents a blanket local-quality claim.

7.2 Interpretation of RQ1

RQ1 produces a negative confirmatory result. Retained thesis TBM is 0.0255 below J-controlled TBM, corresponding to a 6.358% decrease. The development cohort did not predict this direction. The correct interpretation is that development successfully identified unnecessary internal thesis components but did not produce a TBM bank that generalizes beyond the controlled baseline.

Coverage and distinct-PDB selection remain correctly labelled. Coverage is an engineering heuristic selected from development, not a confirmed accuracy term. Distinct-PDB selection prevents spending several slots on the same PDB evidence, but its role is robustness and diversity. Neither item rescues the failed H1 group.

The unsafe-date development diagnostic remains important. Future templates raise the measured score by more than any legitimate development component. Temporal filtering is therefore part of the scientific method, not merely data cleaning, even though the time-safe thesis TBM does not beat J-controlled TBM.

7.3 Interpretation of RQ2

The $3T + 2D$ effect generalizes. Its 9.098% relative TM increase is accompanied by a 28.644% reduction in mean self-TM and a 64.138% reduction in near-duplicate pairs. These native-blind diversity changes provide a coherent mechanism: different sources cover different folds, and best-of-five scoring rewards coverage.

The mechanism is not that DRfold2 is always stronger. The confirmatory $2D$ bank is below $2T$, while $1T + 1D$ is above both. The evidence supports combining independent sources under a fixed budget. It does not support replacing all templates with DRfold2, nor does it prove that DRfold2 is uniquely responsible for the effect.

The absence of a reproducible Boltz source limits the mechanism analysis. A matched Boltz experiment could test whether another independent predictor produces the same effect. Without that comparison, independent-source complementarity is the strongest supported interpretation, while the implementation evidence remains specific to direct DRfold2 candidates.

7.4 Interpretation of RQ3

RQ3 passes on both development and confirmatory data. Its primary evidence is SW-RMSD9, which uses the native structure but is not optimized directly by Geometry. Improvements in angle likelihood, torsion likelihood, clashes or kink count cannot provide the same evidence because these quantities are inside the objective.

The confirmatory SW-RMSD9 reduction is statistically supported but small, 0.473%. Bank TM is 0.532% above Simple and 0.00178 above Raw. Geometry therefore does more than satisfy a noninferiority boundary on this population, although its TM contribution is much smaller than the source-allocation contribution. It is a refiner, not a second fold predictor.

The supporting metrics place an important boundary around this claim. SW-RMSD15 also improves, while global C1' RMSD and candidate TM are inconclusive. C1'-IDDT decreases. Geometry therefore improves the preregistered window-based measure and bank-level TM but does not improve every definition of local structural accuracy.

John fixed further clarifies the trade-off. It gives low windowed error but lowers bank TM. Local correction alone is insufficient. A competitive refiner must be judged jointly by an independent local metric and the global-fold objective.

7.5 Threats to validity

The development cohort contains only 20 previously exposed targets. It is appropriate for component selection but not for confirmatory claims. The separate 97-target set provides stronger evidence about generalization, and cluster-aware inference accounts for 86 sequence-similarity clusters. It does not establish performance on every RNA length, experimental condition or biological class.

The public John reproduction is partial. The original data universe, dependency state, and exact winning submission are unavailable, and the Boltz-conditioned branch cannot be regenerated. The J-controlled TBM comparator remains meaningful for RQ1, but complete-system differences with John are descriptive.

The DRfold2 structural-training screen is stronger than a date-only claim, but RCLM sequence membership is unresolved. The final set cannot establish complete pretrained time-safety. Researcher memory also cannot be independently reconstructed; the external audit can only cover accessible text and artifacts.

The final confirmatory pool uses train-V2 labels that were locally available but kept

unopened for performance until the output freeze and explicit opening receipt. Procedural separation, hashes and a native-blind runner reduce researcher degrees of freedom, but they do not create a truly inaccessible external test server. Kaggle remains the external hidden benchmark for the complete deployed pipeline.

The independent local metrics disagree about Geometry. SW-RMSD9 and SW-RMSD15 improve, but C1'-IDDT decreases. This disagreement is not treated as a technical failure. It indicates that a refiner can improve windowed superposition error while altering pair-distance preservation in an unfavorable way. The thesis therefore avoids a universal geometry-quality claim.

Chapter 8

Reproducibility

The V4 preregistration is frozen at git tag ‘v4-preregistration-2026-08-20’. The master ledger contains 5,135 train-V2 rows, while CASP15 targets are managed separately in the development ledger. Every major artifact records its input path and SHA-256 hash.

The final manifest contains 97 RNAs in 86 clusters. The final freeze records hashes for the manifest, template metadata, template coordinates, P0 priors, DRfold2 checkpoint manifest, pretrained provenance audit, selected refiner configuration, statistical implementation, environment file, TBM pipeline module, and native-blind runner.

The retained TBM implementation is replayed against all 20 frozen development banks. Coordinates, per-residue confidence, and global confidence match exactly for all 20 targets. The replay reads no native coordinates and performs no structural scoring.

The native-blind execution followed this sequence:

1. validate every frozen input and code hash;
2. generate J-controlled and retained thesis template banks;
3. run direct DRfold2 cfg97 on the frozen final manifest;
4. assemble $5T$, $3T + 2D$, and supporting source banks with the frozen fallback;
5. run Raw, John, Simple, and selected Geometry cells without labels;
6. hash the generated outputs and stop;
7. introduce the native evaluator only after an explicit final-opening receipt.

The execution froze 3,011 scientific files totalling 8.940 GB. It produced all 97 TBM, raw and refined banks, 194 selected DRfold2 candidates and zero generation failures. Target-level parallel execution was checked against serial execution on 8BU8_A; all ten refinement

arrays were byte-identical. The receipt then authorized one native-label opening. Evaluation covered all 97 targets with zero scoring failures, and the result files record the unmodified H1, H2, H3 and supporting analyses.

The result ledger is updated immediately after each experiment. Method descriptions, results, and interpretations are refreshed from the same frozen artifact rather than reconstructed from memory. This workflow prevents tables and prose from referring to different runs.

Chapter 9

Conclusion

This thesis uses a competition with a clear best-of-five TM-score objective to study how retrieval, candidate-source allocation, and refinement affect an RNA 3D prediction pipeline. The main methodological contribution is controlled decomposition under shared data and evaluation rules rather than a claim that the complete system was designed independently of the public winning-team work.

The 97-target confirmatory evaluation does not support H1 and instead gives negative evidence. Retained thesis TBM is 0.0255 below J-controlled TBM, a 6.358% decrease. Internal simplification does not translate into a stronger TBM baseline. Coverage must remain a heuristic, distinct-PDB selection a diversity safeguard, and the thesis TBM group an unsuccessful controlled replacement.

H2 passes. Replacing two of five template slots with direct DRfold2 candidates raises mean TM by 0.0342, or 9.098%. Lower self-TM and fewer near-duplicate pairs, together with the matched- N results, support independent-source complementarity. The finding does not mean that DRfold2-only prediction is superior to template-only prediction.

H3 also passes. Selected Geometry reduces SW-RMSD9 beyond Simple by 0.0205 Å, or 0.473%, and raises bank TM by 0.00218, or 0.532%. SW-RMSD15 supports the effect, but global C1' RMSD and candidate TM are inconclusive and C1'-IDDT decreases. Geometry is therefore a small, metric-specific confirmed refinement contribution rather than a universal improvement in local geometry.

The thesis conclusion is deliberately asymmetric. Candidate-source diversification is the main confirmed contribution to the competition TM-score objective. Geometry adds a smaller confirmed effect beyond simple smoothing. The thesis TBM replacement does not generalize and is not defended as a contribution. Unsupported complexity is removed, safeguards are labelled accurately, and negative and conflicting results remain visible. Kaggle is

retained as a complete-system external benchmark and is not used to attribute any component-level effect.

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